

Metabolic Modelling of *Aspergillus niger* and Grapevine Pathogenic Interactions

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Abstract. Viticulture represents a cornerstone of Portugal’s heritage; however, it currently faces significant threats from phytopathogenic fungi such as *Aspergillus niger*. This filamentous fungus acts as an opportunistic pathogen, compromising both crop yield and food safety through the biosynthesis of mycotoxins. Although *Vitis vinifera* deploys innate immune responses, *A. niger* frequently overcomes these barriers, leading to severe metabolic disruption and tissue disintegration. While traditionally managed with chemical fungicides, there is an urgent demand for sustainable alternatives to mitigate environmental impacts. This study employs a systems biology approach, utilizing curated genome-scale metabolic models and flux balance analysis to characterize nutrient cross-feeding and identify critical flux distributions during the infection process. The methodological framework integrates computational tools, such as COBRApy for *in silico* constraint-based simulations — including single-reaction and gene deletion analysis — and Escher for network visualization. By delineating metabolic dependencies and vulnerabilities through the simulation of genetic and nutritional constraints, this research establishes an *in silico* platform to identify essential enzymes and pathways. This theoretical computational foundation facilitates the transition toward a more sustainable viticultural model by pinpointing specific metabolic targets and reducing chemical dependency.

Keywords: FBA · GEMs · *In silico* simulation · Mycotoxins · Plant-fungal interaction · Sustainable agriculture.

1 Overview and Problem Statement

Viticulture constitutes one of the cornerstones of Portugal’s cultural and economic heritage, a status shared by many wine-growing regions globally [1]. The *V. vinifera* species has been cultivated since the Neolithic, with roots deeply embedded in classical Mediterranean civilizations [2]. This long-standing practice has preserved ancestral traditions throughout the production cycle [1]. Economically, its significance transcends direct commercialization; today, the wine sector drives regional development, strengthening the trade balance through exports and stimulating the tertiary sector through wine tourism, which combines enological appreciation with cultural and landscape experiences [3].

Within this framework, any factor threatening viticultural productivity or enological quality emerges as a critical priority for mitigation strategies. Among the main biotic threats, phytopathogenic fungi are particularly notable [4], as they drastically reduce crop yields by inducing vine diseases and compromising food safety through mycotoxin biosynthesis [5]. Disease management traditionally relies on chemical fungicides; however, the environmental persistence of these compounds and their associated ecotoxicological impacts have driven the search for sustainable alternatives [6].

To address these challenges, this project focuses on the computational analysis of metabolic interactions between *V. vinifera* and *A. niger*. Leveraging curated genome-scale metabolic models (GEMs) of both host and pathogen - iMS7199 and iJB1325 -, this research employs constraint-based simulations to characterize the metabolic dependencies sustaining fungal infection. The identification of these key metabolic hubs provides actionable insights for the development of innovative biocontrol strategies. Ultimately, these findings pave the way for a resilient viticultural framework that balances crop protection with environmental management [7].

1.1 Objectives

The primary goal of this research is to develop a computational framework to analyze the metabolic interactions between *V. vinifera* and *A. niger*, focusing on the identification of key metabolic dependencies through constraint-based simulations. To ensure a feasible and focused study, the following specific objectives have been defined:

- **Curation and Functional Validation of Metabolic Networks:** Perform a targeted curation of the GEMs for *V. vinifera* (iMS7199) and *A. niger* (iJB1325). This involves ensuring basic stoichiometric consistency and resolving essential mass-balance errors and blocked reactions using the Memote testing suite to guarantee model reliability.
- **Characterization of Interspecific Nutrient Cross-feeding:** Develop a simplified interaction model to identify a specific set of potential metabolites — primarily sugars and organic acids — exported by the host and utilized by *A. niger* for biomass production. This involves establishing unidirectional transport fluxes to quantify the metabolic interaction potential (MIP) within a compartmentalized framework, utilizing Escher for the spatial visualization of these interspecific metabolic exchanges.
- **Identification of Fungal Metabolic Vulnerabilities:** Conduct single-gene and single-reaction deletion analyses using COBRApy to identify essential metabolic hubs required for *A. niger* growth under host-simulated nutritional constraints. These targets will be evaluated based on their impact on fungal fitness and screened for homology against the host genome to ensure target specificity, identifying theoretical intervention points for future sustainable biocontrol strategies.

2 State of the Art

2.1 The Host: *Vitis vinifera*

The biological identity of *V. vinifera* is defined by a relatively compact genome of approximately 494.9 Megabase pairs (Mbp), organized into 19 pairs of chromosomes ($2n = 38$) [8]. Although it was the first fruit species to have its genome sequenced, its structural complexity remains high due to pronounced heterozygosity, repetitive sequences, and substantial intraspecific variability, including copy number variations and single nucleotide polymorphisms between cultivars [9]. Recent telomere-to-telomere assemblies have enabled complete mapping of telomeres and centromeres, providing crucial tools for genetic improvement, identification of disease resistance genes, and adaptation to climate change [10].

From a systems biology perspective, the plant's development is governed by a highly regulated metabolic network that is extremely sensitive to environmental conditions [11] and dictates the metabolic state of the berry. For the purpose of metabolic modelling, the ripening stage (*véraison*) is the most critical, as it is characterized by a significant metabolic shift: the accumulation of soluble solids (sugars) and a decrease in organic acids [12,13,14]. This period defines the nutritional landscape available to phytopathogens.

The grapevine's metabolic system is categorized into two primary axes that facilitate the systems biology analysis:

- **Primary Metabolism:** Occurs mainly in leaves and fruits. Functions as the core source of essential precursors for the interaction model, providing carbohydrates, organic acids (tartaric and malic), and amino acids [15] that serve as inputs for the fungal metabolic network, the primary carbon and nitrogen sources for fungal growth.
- **Secondary Metabolism:** Comprises the synthesis of bioactive compounds involved in defense mechanisms — such as phenolic compounds [16], stilbenes and phytoalexins (e.g. trans-resveratrol and ϵ -viniferin [17]) — which act as biochemical constraints and inhibitory factors during host-pathogen metabolic simulations.

2.2 The Pathogen: *Aspergillus niger*

As one of the most common and geographically widespread species within its genus, *A. niger* is a filamentous saprophyte renowned for its remarkable metabolic versatility and its role as a high-efficiency "cellular factory" [18]. The genome of *A. niger* spans approximately 34 Mbp, is organized into 8 chromosomes ($n = 8$), and contains roughly 14,532 genes encoding 14,086 proteins [19]. Genomic analyses revealed a vast repertoire of hydrolytic and degradative enzymes for polysaccharide breakdown [20]. From a systems biology perspective, this genetic potential translates into a robust metabolic network capable of diverse biochemical transformations, making it an ideal candidate for genome-scale modeling.

From a metabolic modelling perspective, the pathogenicity of *A. niger* is intrinsically linked to its high-flux primary and secondary metabolism. The metabolic network is structured into two main functional outputs:

- **Primary Metabolism:** Characterized by a high flux toward the synthesis of organic acids (notably citric and gluconic acid) and the secretion of a wide range of extracellular enzymes, such as amylases and proteases [21], which facilitate the degradation of host-derived substrates in the interaction model.
- **Secondary Metabolism:** Responsible for the biosynthesis of polyketides and alkaloids, including mycotoxins such as fumonisins and ochratoxin A (OTA) [22], which pose nephrotoxic and carcinogenic risks. In the context of flux balance analysis (FBA), these pathways represent critical metabolic sinks that influence the overall energy homeostasis and virulence potential of the fungus during infection.

Although it is a strictly aerobic saprophyte, the fungus’s ability to persist in diverse environments is supported by its tolerance to environmental stress and a highly adaptable oxidative response, factors that are integrated into the metabolic model as specific survival constraints [23,24].

2.3 Host-Pathogen Interaction

Within the spectrum of phytopathogens that affect *V. vinifera*, *A. niger* represents a notable threat, acting primarily from ripening through harvest [25]. The infection process is characterized by secretion of cell wall-degrading enzymes [26] and accumulation of organic acids [27,28], leading to tissue disintegration.

From a systems biology perspective, this interaction is a dynamic shift in the host’s metabolic homeostasis. While *V. vinifera* activates innate immune responses — such as the synthesis of fungitoxic stilbenes and phytoalexins (*trans*-resveratrol [29] and ϵ -viniferins) and the induction of defense genes — *A. niger* often exhibits sufficient resilience to overcome these barriers [30].

For the purpose of constraint-based modelling, the most critical aspect of this interaction is the severe metabolic disruption induced by the pathogen:

- **Energy Homeostasis:** The infection causes a significant downregulation of glycolysis and the tricarboxylic acid cycle in the berry, compromising adenosine triphosphate production [31,32].
- **Nutritional Favorability:** As the fruit ripens, increase in sugar levels and reduction in acidity optimize conditions for fungal flux distribution [33].
- **Virulence Factors:** The production of mycotoxins like OTA and fumonisins by *A. niger* interferes with the plant’s detoxification pathways and redox balance, providing additional constraints that favor fungal fitness over host resilience, overcoming the host resistance and resulting in lasting structural damage [34,35].

This metabolic interface can be represented as a compartmentalized system where the trade-offs between host defense and pathogen growth are quantified through flux distributions.

3 Methodology

3.1 Metabolic Modelling and Constraint-Based Analysis

Integration of multi-omics data into a functional understanding of complex biological systems is achieved through GEMs [36,37]. These models serve as comprehensive biochemical scaffolds for the metabolic potential of an organism, enabling *in silico* simulation of intracellular flux distributions under various conditions.

The development of GEMs for grapevine has advanced significantly, with models such as **iMS7199** [38] providing a robust platform to study complex interactions between primary and secondary metabolism. Resources such as **VitisCyc** [39] and **VitisNet** [40] allow for detailed exploration of metabolic pathways, facilitating the integration of omics data into network-level analyses.

Similarly, *A. niger* research has benefited from refined GEMs, such as the consensus model **iJB1325** [41], a reference for simulating high-flux organic acid biosynthesis, reconstructed for the reference strains **CBS 513.88** and **ATCC 1015**. These models have been instrumental in studying industrial enzyme production and organic acid biosynthesis [42]. However, a significant knowledge gap persists regarding the dynamic metabolic exchanges that occur during active infection of *V. vinifera*, particularly concerning the trade-offs between fungal fitness and the biosynthesis of virulence factors.

The technical core of this research relies on Constraint-Based Reconstruction and Analysis (COBRA). This framework mathematically represents the metabolic network as a stoichiometric matrix ($S_{m \times n}$), where rows represent m metabolites and columns represent n reactions. Assuming a pseudo-steady state ($S \cdot v = b$, where v is the vector of metabolic fluxes and b represents the rate of change of metabolite concentrations, here assumed to be zero), FBA [43] is employed to solve an optimization problem, typically maximizing the biomass objective function ($Z = c^T v$) under specific environmental constraints.

To ensure the reliability of the simulations, this study integrates a quality control workflow:

- **Model Validation:** The GEMs are validated through Memote [44], a standardized testing suite that assesses stoichiometric consistency, mass-charge balance, and the presence of blocked reactions, ensuring the models meet the community’s quality standards.
- **Compartmentalized Modelling:** Rather than automated pipelines, a direct compartmentalized approach is used to integrate host and pathogen into a unified metabolic landscape. This allows for the quantification of the MIP by establishing specific transport reactions that define the nutrient cross-feeding interface.
- **Computational Simulation:** The implementation is performed using COBRApy [45]. The environmental constraints, specifically the lower and upper bounds for nutrient uptake reactions (e.g., glucose, fructose, and organic acids) and oxygen exchange, are parameterized based on the physiological data of the berry during the véraison stage, as characterized previously.

Single-gene and single-reaction deletion analyses are utilized to identify essential metabolic hubs — critical nodes where the loss of function significantly reduces fungal fitness.

- **Visualization:** The resulting flux distributions and interaction networks are mapped using Escher [46], providing a clear interpretation of the metabolic dependencies during infection.

The methodological strategy implemented in this study is detailed in Fig. 1, which systematizes the transition from the preparation of GEMs to the identification of fungal metabolic vulnerabilities.

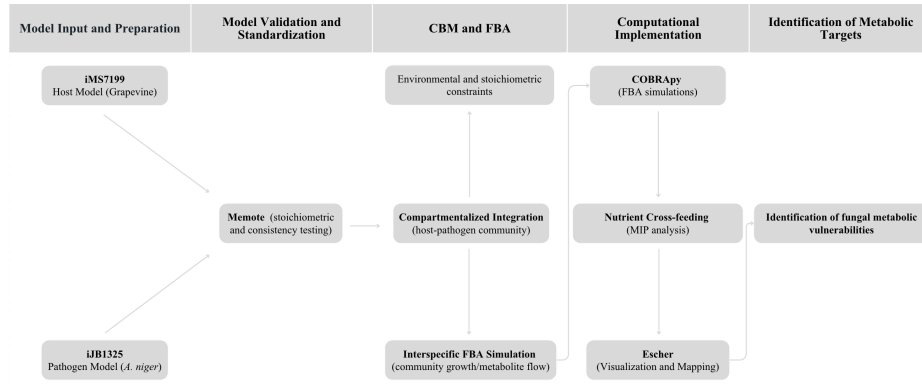


Fig. 1. Methodological workflow for the systems biology analysis of Grapevine – *A. niger* interspecific metabolic interactions, encompassing model validation, compartmentalized constraint-based simulation, and *in silico* identification of fungal metabolic vulnerabilities (elaborated by the author).

By integrating these computational frameworks, this study provides a predictive map of the metabolic interface, establishing a theoretical foundation that could, in the future, support sustainable agricultural applications and the development of innovative biocontrol strategies.

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